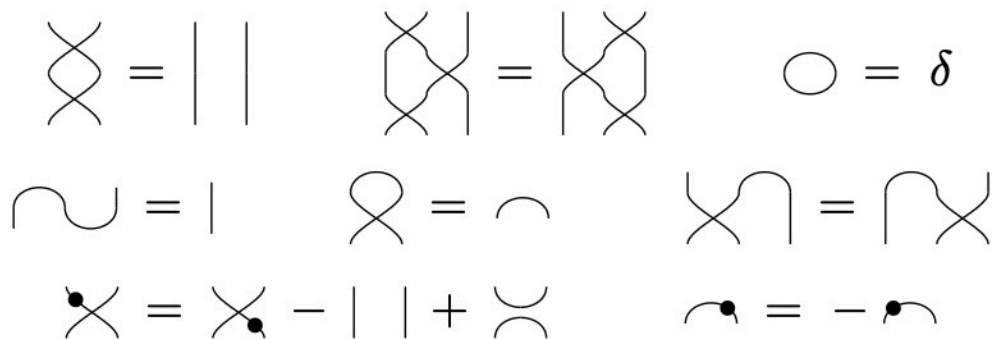


The affine Brauer category AB has generators:



and relations:



+ more rules ($X^\bullet = \bullet X + 11 - \cup$) and conditions

MONOIDAL GRÖBNER SYSTEMS & CATEGORIES OF AFFINE BRAUER TYPE

jr Sigiswald Barbier

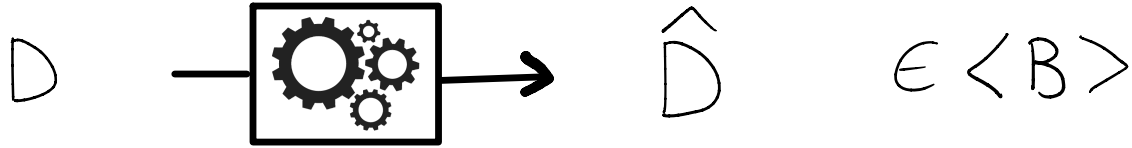
tensor categories in representation theory & vice-versa

Uppsala, 26 August 2026



Léo Schelstraete
Max Planck Institute for Mathematics

Q How to show that a presented linear monoidal category has the expected hom-basis?



B defines a basis $\Leftrightarrow$ algorithm gives unique output

assoc. alg.: Gröbner-Shirshov bases / diamond lemma (60s-70s)

$\leadsto$ underlying algebra software

$\leadsto$ instances of "rewriting systems"

A monoidal Gröbner systems (S.-Barbier 26+, based on S. 25)

A category of affine Brauer type is a linear monoidal category generated by



st any set of representatives of dotted reduced matchings gives a basis.

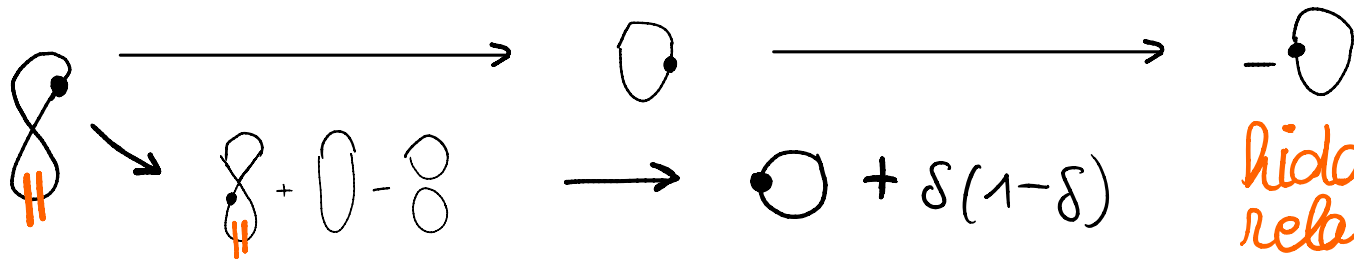
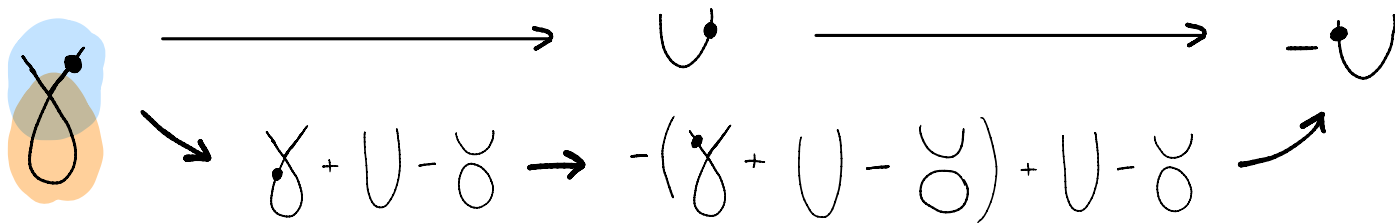
A category of affine Brauer type is a linear monoidal category generated by



st any set of representatives of dotted reduced matchings gives a basis.

= "same underlying hom-spaces as AB "

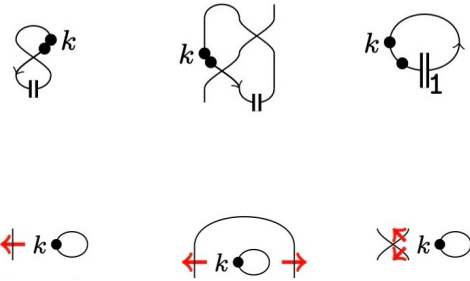
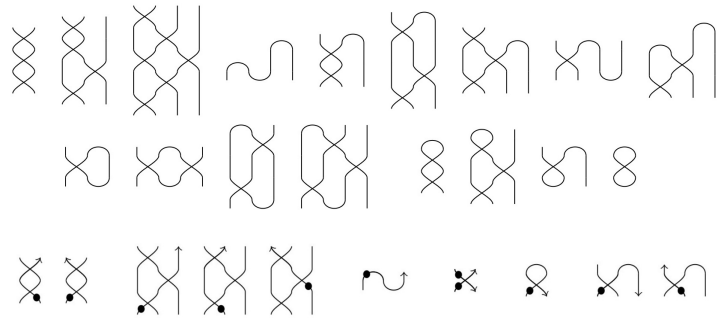
→ + choice of bubbles



hidden relation!

Main Theorem* (S.-Barbier 26+)

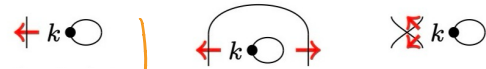
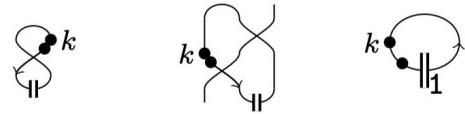
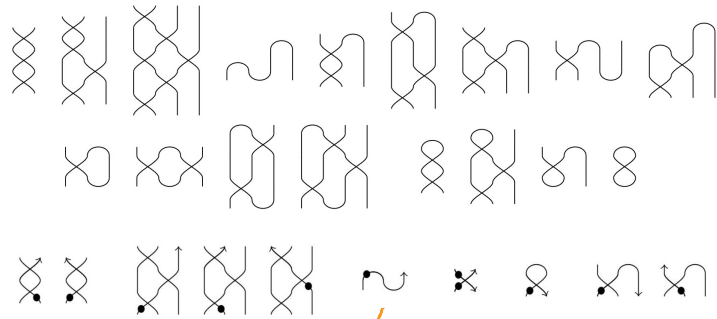
Let $\mathcal{C}$ be presented by a rewriting system of affine Brauer type.
 If each of the following diagrams has a unique normal form:



then $\mathcal{C}$ is of affine Brauer type.

Main Theorem* (S.-Barbier 26+)

Let $\mathcal{C}$ be presented by a rewriting system of affine Brauer type.
If each of the following diagrams have a unique normal form:



then $\mathcal{C}$ is of affine Brauer type.

implemented

global termination condition!

generating functions

Examples

Known examples:

- affine Brauer AB_δ [RS23] $2N$ bubbles
- affine VW [BDE-AHHILNSS18] $\emptyset$ bubbles
- nil-Brauer NB_δ [BW25] $2N+1$ bubbles

Theorem (BW26) NB_δ categorifies the split quantum group of rank 1.

The odd nil-Brauer category ($\delta \in \{0, -1\}$, $p(v) = \delta + 1 \pmod{2}$)

$$\times$$

odd

•
odd

$$\text{Diagram of two crossing lines} = 0$$

A diagrammatic equation showing the crossing of two lines. On the left, two vertical lines cross each other. On the right, the same two lines are shown, but they are now separated by a small gap, with an equals sign between the two diagrams.

$$\bigcirc = \delta$$

$$\text{[Diagram: A wavy line with a peak and a trough, followed by an equals sign and a vertical line.]}$$

$$\text{figure-eight} = 0$$

A diagrammatic equation showing the crossing of two strands. On the left, two vertical strands cross each other. On the right, the same two strands are shown, but they are connected by a horizontal arc at the top, forming a loop. The two diagrams are separated by an equals sign.

$$p(\cup)$$

$$p(\cup)$$

$$\text{Diagram 1} = - \text{Diagram 2} - \text{Diagram 3} - (-1)^{p(\cup)} \text{Diagram 4}$$

$$\text{cap} = -(-1)^{p(\cup)} \text{cap}$$

Theorem (S-Barbier 26) ONB_8 is a category of affine Brauer type with $2N+1$ bubbles.

The odd nil-Brauer category ($\delta \in \{0, -1\}$, $p(v) = \delta + 1 \pmod{2}$)

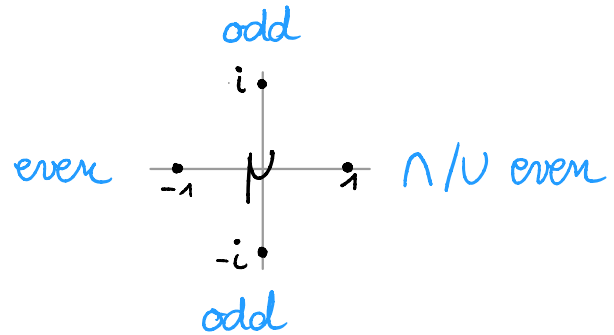
$$\begin{array}{ccccccc}
\begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} & \begin{array}{c} \bullet \\ | \end{array} & & \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} = 0 & \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} = \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} & & \bigcirc = \delta \\
\textit{odd} & \textit{odd} & & & & & \\
\begin{array}{c} \cup \end{array} & \begin{array}{c} \cup \end{array} & & \begin{array}{c} \cup \end{array} = | & \begin{array}{c} \cup \end{array} = 0 & & \begin{array}{c} \cup \end{array} = \begin{array}{c} \cup \end{array} \\
p(\cup) & p(\cup) & & & & & \\
\begin{array}{c} \bullet \\ \diagup \diagdown \\ \diagdown \diagup \end{array} = - \begin{array}{c} \diagup \diagdown \\ \bullet \\ \diagdown \diagup \end{array} - | \quad | - (-1)^{p(\cup)} \begin{array}{c} \cup \end{array} & & \begin{array}{c} \cup \bullet \end{array} = -(-1)^{p(\cup)} \begin{array}{c} \bullet \cup \end{array}
\end{array}$$

Theorem (S-Barbier 26) ONB_8 is a category of affine Brauer type with $2N+1$ bubbles.

Q: categorify super quantum group?

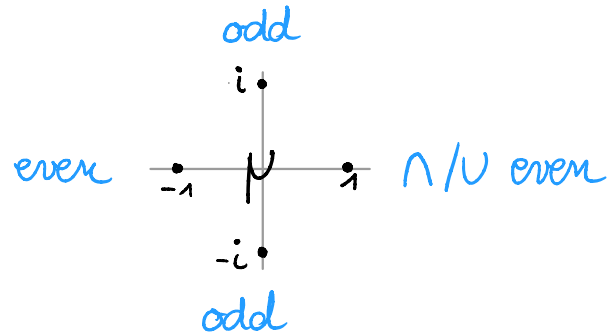
The twisted affine Brauer AB^p ($\mu^4 = 1$) :

$$\begin{array}{lll}
 \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} = || & \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} = \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} & \bigcirc = \delta \\
 \cup = | & \cap = \cap & \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} = \mu \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} \\
 \begin{array}{c} \bullet \\ \times \end{array} = \begin{array}{c} \times \\ \bullet \end{array} - || + \mu^{-1} \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} & \begin{array}{c} \cap \\ \bullet \end{array} = - \begin{array}{c} \bullet \\ \cap \end{array}
 \end{array}$$



The twisted affine Brauer AB^p ($\mu^4 = 1$):

$$\begin{aligned}
 \text{X} &= || & \text{X} &= \text{X} & \bigcirc &= \delta \\
 \cup &= | & \cap &= \cap & \text{X} &= \mu \cap \\
 \times &= \times - || + \mu^{-1} \cap & \cap &= - \times
 \end{aligned}$$



$\leadsto$ parity depends on μ !

$\leadsto AB^p \cong AB$ as linear categories

Theorem (S-Barbier 26+) Let $\mathcal{C}$ presented by:

$$\begin{array}{lll}
 \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} = d \begin{array}{|l} \hline \\ \hline \end{array} & \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} = \lambda \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} & \bigcirc = \delta \\
 \smile = \zeta \begin{array}{|l} \hline \\ \hline \end{array} & \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} = a \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} & \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} = \mu \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} \\
 \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} = \alpha \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} + b \begin{array}{|l} \hline \\ \hline \end{array} + c \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} & \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} = \sigma \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} + \tau \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array}
 \end{array}$$

If $\mathcal{C}$ is a category of affine Brauer type, then $\mathcal{C}$ is isomorphic, as a linear category, to AB_δ , VW , NB_δ or ONB_δ .

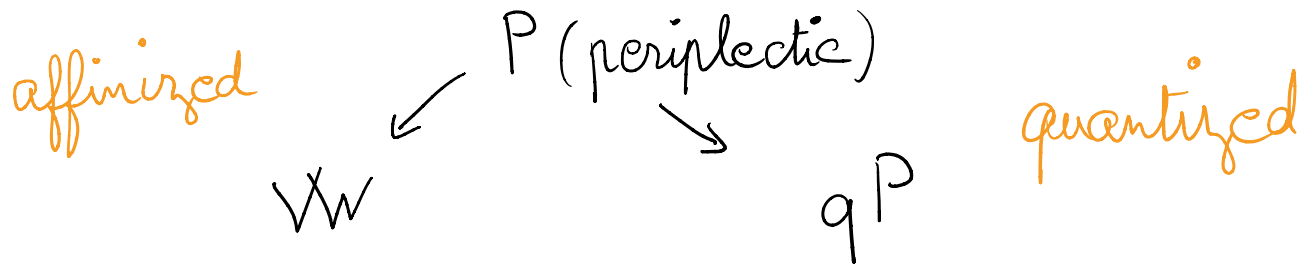
Theorem (S-Barbier 26+) Let $\mathcal{C}$ presented by:

$$\begin{aligned}
 \text{X} &= d \mid \mid & \text{X} &= \lambda \text{X} & \bigcirc &= \delta \\
 \smile &= \zeta \mid & \bowtie &= a \smile & \text{X} &= \mu \text{X} \\
 \text{X} &= \alpha \text{X} + b \mid \mid + c \text{X} & \smile &= \sigma \smile + \tau \smile
 \end{aligned}$$

invertible

If $\mathcal{C}$ is a category of affine Brauer type, then $\mathcal{C}$ is isomorphic, as a linear category, to AB_δ , VW , NB_δ or ONB_δ .

$\leadsto$ also a monoidal classification



The quantized affine VW category:

$$\begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} = | \quad | - (q - q^{-1}) \times \quad \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} = \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} \quad \text{AND}$$

$$\cup = | \quad \cap = -q \cup \quad \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} = \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} - (q - q^{-1}) | \quad \cap$$

$$\bullet \times = \times \bullet + q^{-1} | \quad | - (q - q^{-1}) \bullet \quad | + q^{-3} \smile + (q^{-3} - q^{-1}) \smile \bullet$$

$$\smile \bullet = q^{-2} \bullet \smile + q^{-2} \cap \quad \bullet \bigcirc_k = 0$$

Future directions:

- interpret new examples
- construct new examples / classification
- oriented Brauer? Kac-Moody 2-categories?
- other diagrammatics?

THANK YOU

Theorem 1.12. Fix a field $\mathbb{k}$ where 2 is invertible. Let $\mathcal{C}$ be a supermonoidal category over $\mathbb{k}$ presented by an airy presentation of affine Brauer type (Definition 1.9). Assume further that either b or c are invertible. Then $\mathcal{C}$ is a category of affine Brauer type if and only if, up to isomorphism of filtered supermonoidal categories (where the filtration is the filtration by the number of dots), the category $\mathcal{C}$ admits a presentation of the form

$$\begin{array}{c}
\begin{array}{ccc}
\begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} = \mathbf{a} \begin{array}{|c|} \hline | \\ \hline \end{array} & \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} = \lambda \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} & \bigcirc = \delta \\
\begin{array}{c} \smile \\ \frown \end{array} = \begin{array}{|c|} \hline | \\ \hline \end{array} & \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} = \mathbf{a} \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} & \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} = \mu \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} \\
\begin{array}{c} \bullet \diagup \diagdown \\ \diagdown \diagup \bullet \end{array} = \alpha \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} - \begin{array}{|c|} \hline | \\ \hline \end{array} + c \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} & \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} = \sigma \begin{array}{c} \bullet \diagup \diagdown \\ \diagdown \diagup \bullet \end{array} + \tau \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array}
\end{array}
\end{array}$$

such that $p(\cup) = p(\cap)$ and the remaining parities and parameters are as follows:

$\begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} \mathcal{B} \downarrow$	$\mathbf{x}$	δ	$\mathbf{a}$	λ	μ^2	α	c	σ	τ	<i>e.g.</i>
<i>even</i>	$2\mathbb{N}_{>}$	δ	1	1	$(-1)^{p(\cup)}$	1	μ^{-1}	-1	0	$\mathcal{AB}_{\delta}$
<i>even</i>	$\emptyset$	0	1	1	$-(-1)^{p(\cup)}$	1	$-\mu^{-1}$	1	1	$\mathcal{W}$
<i>even</i>	$2\mathbb{N} + 1$	$\delta \in \{0, \lambda\mu\}$	0	$\lambda^2 = 1$	$(-1)^{p(\cup)}$	λ	$\lambda\mu^{-1}$	$-\lambda$	0	$\mathcal{NB}_{\delta}$
<i>odd</i>	$2\mathbb{N} + 1$	$\delta = -\lambda\mu$	0	$\lambda^2 = 1$	$(-1)^{p(\cup)}$	$-\lambda$	$-\lambda\mu^{-1}$	$-\lambda$	0	$\mathcal{ONB}_{-1}$
<i>odd</i>	$2\mathbb{N} + 1$	$\delta = 0$	0	$\lambda^2 = 1$	$-(-1)^{p(\cup)}$	$-\lambda$	$\lambda\mu^{-1}$	λ	0	$\mathcal{ONB}_0$